ELG5378:
IMAGE PROCESSING AND IMAGE COMMUNICATION
WINTER 2026
TERM PROJECT

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The term project will be finished in a group of two (2) students. Each group will choose a topic from Section 3, prepare required knowledge by reading research papers, and complete the project.

1. Paper Reading

Each group is required to read one (1) to three (3) journal or conference papers that represent the current development of the chosen area (in Section 3). Normally, a group will fulfil the research and implementation based on these papers. Each group will be allocated 15-20 minutes to present the paper(s) (including the implementation results they achieved) to the class, which will be scheduled in the last two weeks.

2. Term Project

This project should address a subject related to image processing and image communication. The goal of the project is to present a specific application of image processing techniques to a practical problem based on a textbook or one or more articles in the literature. It is required to describe the application and to present the mathematical description of the solution. It is desirable to use the terminology and notation introduced in the course as much as possible. An experimental component involving simulation (possibly simplified) of the problem and of the method is also required. I expect you to do new work for this project (i.e., do not resubmit work you have done before for some other purpose, or for another course project at the same time). You may make use, in part, of public domain (or other) software packages, MATLAB, etc., as long as you clearly state this. A short progress report of at most five pages, indicating what has been accomplished so far, should be submitted. The final report should be at most 20 pages and be prepared with a text-processing system, preferably LaTeX.

The choice of subject should be made in the next two weeks. A short description of the problem selected and several pertinent references (1-2 pages) should be submitted to the professor (BrightSpace). You must write the report yourself; plagiarism can lead to a failing mark in the project, the course, or worse. See the leaflet "Academic Integrity" attached to the end of this file for "How to Avoid Plagiarism"; please read it carefully and include a statement in your proposal that you have read and understood the leaflet.

Date: Posted on January 22, 2026.

Below is a list of a few sample topics. You may suggest other related topics of particular interest to you.

3. Possible Topics

- Image restoration
- Color imaging
- Edge detection
- Image panoramas
- Image enhancement
- Stereoscopic imaging
- Image interpolation, scan conversion
- 2D FIR filter design
- Lossless image compression
- Nonlinear filtering
- Document compression
- Texture modeling and synthesis
- Still image compression
- Video compression
- Motion estimation
- Disparity estimation
- Computational photography
- Image keying
- Image watermarking
- Depth map compression
- Image quality evaluation
- Super-resolution
- Video de-interlacing
- Video super-slow-motion
- Video frame interpolation
- Application of deep learning on image processing

Ideas for topics can be found in recent issues of IEEE Transactions on Image Processing, IEEE Transactions on Circuits and Systems for Video Technology, Signal Processing: Image Communication, and in the Proceedings of the International Conference on Image Processing.

4. Tentative Schedule for Project

Date	Submission	Length	Weight
02/05/2026	Project proposal	1-2 pages	10 %
03/04/2026	Progress report	3-5 pages	10 %
TBD	Paper presentation	15-20 minutes	30 %
04/30/2026	Final project report	20 pages at most	50 %

University of Ottawa

Academic Integrity

Student's Guide

uOttawa

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Canada's university

Academic Integrity

Student's Guide

Academic integrity means being responsible for the quality of your work, preparing it honestly and respecting the intellectual community you are part of as a student. It is a core value in all scholarly work.

Academic fraud refers to "an act by a student that may result in a false academic evaluation of that student or of another student" *(Policy on Academic Fraud)*. Here are some examples:

- Submitting work prepared by someone else or for someone else
- Using work you have previously submitted for another course, without your professor's permission
- Falsifying or making up information or data
- Falsifying an academic evaluation
- Submitting work you have purchased on the Internet
- Plagiarizing (see below) ideas or facts from others

Plagiarism means using words, sentences, ideas and facts you have gotten from others and passing them off as yours, by failing to quote or reference them correctly. Plagiarism comes in many forms, including the following:

- Failing to place words or sentences you have taken from other authors in quotation marks ("…")
- "Copying and pasting" information found on the Internet without providing a reference
- Translating texts without providing a reference for their sources
- Not providing a reference for a paraphrase or a summary

How to Avoid Plagiarism

Since your academic work must be based on research, you need to learn the proper way to quote, paraphrase and summarize, in order to be sure you are not plagiarizing.

Quoting means using another author's exact words in your text. The citation must be placed in quotation marks ("…") in your text, using the appropriate bibliographical referencing style (see resources below). Websites, pictures, statistics, graphs and tables taken from another source must be referenced.

~ Referencing styles may vary according to faculty, discipline and even professor; ask your professor which style you should use (for example, APA, MLA, Chicago Manual).

Example

Source text: Over time technology has been instrumental in increasing industrial and agricultural production, improving transportation and communications, advancing human health care and overall improving many aspects of human life. However, much of its success is based on the availability of land, water, energy, and biological resources of the earth.

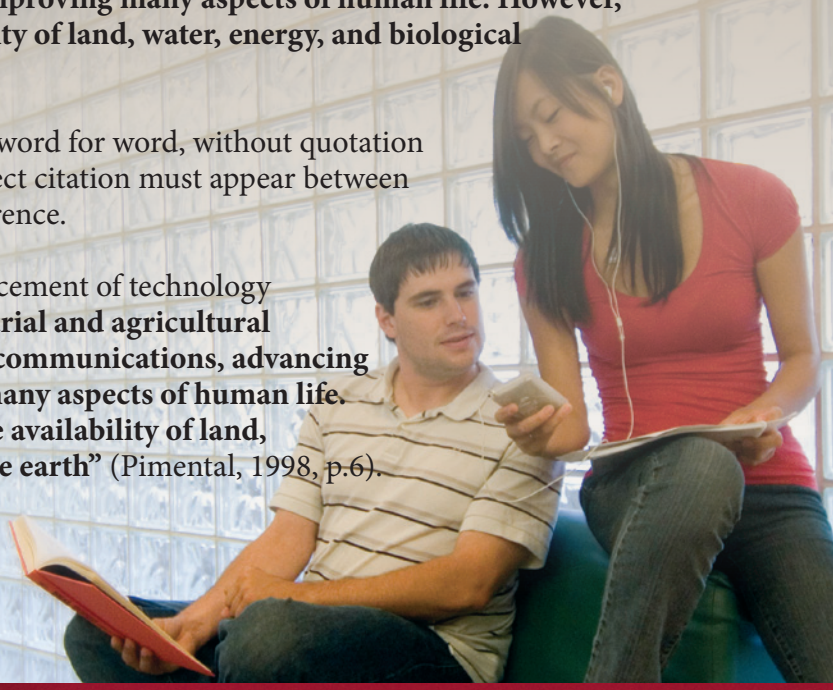
Pimental, David. (1998). Population growth and the environment: Planetary stewardship. *Electronic Green Journal, 1(9).* Retrieved from http://repositories.cdlib.org/uclalib/egj/vol1/iss9/art10

You wrote: Research has shown that **technology has been instrumental in increasing industrial and agricultural production, improving transportation and communications, advancing human health care and overall improving many aspects of human life. However, much of its success is based on the availability of land, water, energy, and biological resources of the earth.***

Problem: Most of the text has been recopied word for word, without quotation marks or any indication of the source. A correct citation must appear between quotation marks in the text, along with a reference.

Solution : Research has shown that the advancement of technology **"has been instrumental in increasing industrial and agricultural production, improving transportation and communications, advancing human health care and overall improving many aspects of human life. However, much of its success is based on the availability of land, water, energy, and biological resources of the earth"** (Pimental, 1998, p.6).

* Words in bold are from the source text.



Paraphrasing means putting into your own words facts, ideas or words from another author.

~ Simply replacing words with their synonyms or changing the sentence structure is not sufficient when you are paraphrasing.

Example

Source text: Over time technology has been instrumental in increasing industrial and agricultural production, improving transportation and communications, advancing human health care and overall improving many aspects of human life. However, much of its success is based on the availability of land, water, energy, and biological resources of the earth.

Pimental, David. (1998). Population growth and the environment: Planetary stewardship. *Electronic Green Journal*, 1(9). Retrieved from http://repositories.cdlib.org/uclalib/egj/vol1/iss9/art10

You wrote: Over time the advancement of technology has been beneficial to the areas of agricultural and industrial production and communication and transportation fields. Furthermore, science has greatly improved health care and many other facets of human life. Nevertheless, its success depends on the availability of natural resources.

Problem: Although some words have been changed, the ideas are borrowed and there is no indication of the source. You cannot simply replace words with their synonyms - this is not paraphrasing. To paraphrase an author's ideas correctly, you must rewrite the ideas using your own words and your own writing style.

Solution : According to Pimental (1998), technology has greatly improved our standard of living. He cautions, however, that technological progress is dependent on natural resources.

Summarizing means taking a section of another author's text or the key elements of an author's ideas and using your own words to present this work in a shorter, concise form.

~ Preserve the meaning of the larger text in a more concise form.

~ Respect what the original author wants to say and present the ideas in the same order.

Example

Source text: But the firm that Walton started has a key asset, the 100 million US consumers who weekly go in search of its "everyday low prices". Low they certainly are, averaging 14% less than the competition (5). The big question is: what is the real cost of these low prices? The answer depends on whether you are concerned with individual buyers looking for the best deal, or with the employees of thousands of suppliers that are in thrall to a company powerful enough to oblige each supplier to hold down, and even reduce, its costs. Workers suffer for the good of Wal-Mart customers. To keep prices rock-bottom in the stores and at the subcontractors, working conditions can only deteriorate. It is consequently easier for suppliers with no unions or for goods manufactured in China (see *"Slaves of the stacked shelves"*).

Halimi, Serge. (2006). Wal-Mart, the new southern plantation. *Le Monde Diplomatique*, 1(5). Retrieved from http://mondediplo.com/2006/01/05walmart

You wrote: Halimi explains that Wal-Mart maintains its low prices by pressuring the producers and enabling bad work conditions on the supply end.

Problem: The summary is too concise and thus it does not preserve the full meaning of the original text.

Solution: Halimi believes that Wal-Mart's 100 million US consumers are captive to its low prices, but they are blind to the fact that those prices are possible only because the company is pressuring the producers and fostering poor work conditions.

How Can You Show Academic Integrity in These Situations?

Authorized group work – When group work is authorized by your professor, all group members contribute to the final product and everyone's contribution should be acknowledged. The workload should be fairly distributed. Each member is responsible for the integrity of the final product and should therefore be sure to read the final version, save a copy and check the referencing of sources used in all sections of the work.

Individual work – When you are working individually, it is important to avoid falling into the plagiarism trap. You must therefore properly document all your source material and properly reference it. Never share your individual work with anyone else, because you run the risk of being accused of plagiarism if part of your work is copied, with or without your permission, and submitted by another student.

Written tests or exams – Remember that helping a fellow student during a test or an exam by sharing answers is clearly cheating. Be careful during exams, and let the professor or proctor know if you see that a student is copying your answers.

The following resources were used in developing the material for this brochure:

The Academic Integrity Website has useful resources for professors and students

Says Who? Integrity in Writing: Avoiding Plagiarism

Beware of Plagiarism! It's Easy, It's Tempting… But It Can Be Very Costly!

Professor's manual for the course SCS1550

Other useful resources

Writing and Style Guide for University Papers and Assignments

The Academic Writing Help Centre's Writing Kit (a guide prepared by SASS)

Reference guides

- APA (American Psychological Association) style
- CM (Chicago Manual) style
- MLA (Modern Language Association) style